

① Solve the following system by using Gaussian-Elimination method:

$$2x_1 + 2x_2 + 2x_3 = 0$$

$$-2x_1 + 5x_2 + 2x_3 = 1$$

$$8x_1 + x_2 + 4x_3 = -1$$

Solⁿ: The augmented matrix is,

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array} \right] R_1' = R_1/2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array} \right] \begin{array}{l} R_2' = R_2 + 2R_1 \\ R_3' = R_3 - 8R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array} \right] R_3' = (-1)R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & 7 & 4 & 1 \end{array} \right] R_3' = R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] R_2' = R_2/7$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

~~Let~~

The above procedure produces a row echelon form and is called Gaussian-Elimination.

The corresponding system of linear equation is,

$$x_1 + x_2 + x_3 = 0 \quad \text{--- (i)}$$

$$x_2 + \frac{4}{7}x_3 = \frac{1}{7} \quad \text{--- (ii)}$$

Let, $x_3 = t$

$$(ii) \Rightarrow x_2 + \frac{4}{7}t = \frac{1}{7}$$

$$\Rightarrow x_2 = \frac{1}{7} - \frac{4}{7}t$$

$$(i) \Rightarrow x_1 + \frac{1}{7} - \frac{4}{7}t + t = 0$$

$$\Rightarrow x_1 = -\frac{1}{7} + \frac{4}{7}t - t$$

$$= \frac{3t}{7}$$

$$= -\frac{1}{7} + \frac{4t-7t}{7}$$

$$= -\frac{1}{7} - \frac{3}{7}t$$

Thus the general soln is,

$$(x_1, x_2, x_3) = \left(-\frac{1}{7} - \frac{3}{7}t, \frac{1}{7} - \frac{4}{7}t, t\right)$$

Ans

(2) Solve the system by using Gaussian elimination

$$\begin{aligned}x_1 + 2x_2 - x_3 + x_4 &= 6 \\-x_1 + x_2 + 2x_3 - x_4 &= 3 \\2x_1 - x_2 + 2x_3 + 2x_4 &= 14 \\x_1 + x_2 - x_3 + 2x_4 &= 8\end{aligned}$$

Soln:

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 6 \\ -1 & 1 & 2 & -1 & 3 \\ 2 & -1 & 2 & 2 & 14 \\ 1 & 1 & -1 & 2 & 8 \end{array} \right] \begin{array}{l} R_2' = R_1 + R_2 \\ R_3' = R_3 - 2R_1 \\ R_4' = R_4 + R_2 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 6 \\ 0 & 3 & 1 & 0 & 9 \\ 0 & -5 & 4 & 0 & 2 \\ 0 & 2 & 1 & 1 & 11 \end{array} \right] \begin{array}{l} R_3' = 3R_3 + 5R_2 \\ R_4' = 3R_4 - 2R_2 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 6 \\ 0 & 3 & 1 & 0 & 9 \\ 0 & 0 & 17 & 0 & 51 \\ 0 & 0 & 1 & 3 & 15 \end{array} \right] \begin{array}{l} R_4' = 17R_4 - R_3 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 6 \\ 0 & 3 & 1 & 0 & 9 \\ 0 & 0 & 17 & 0 & 51 \\ 0 & 0 & 0 & 51 & 170 \end{array} \right] \begin{array}{l} R_2' = R_2/3 \\ R_3' = R_3/17 \\ R_4' = R_4/51 \end{array}$$

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 6 \\ 0 & 1 & \frac{1}{3} & 0 & 3 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{array} \right]$$

The corresponding system of linear equation is,

$$x_1 + 2x_2 - x_3 + x_4 = 6 \quad \text{--- (1)}$$

$$x_2 + \frac{1}{3}x_3 = 3 \quad \text{--- (2)}$$

$$x_3 = 3 \quad \text{--- (3)}$$

$$x_4 = 4 \quad \text{--- (4)}$$

$$(2) \Rightarrow x_2 + 1 = 3$$

$$\Rightarrow x_2 = 2$$

$$(1) \quad x_1 + 4 - 3 + 4 = 6$$

$$\Rightarrow x_1 = 6 - 5 = 1$$

Thus the general solⁿ is

$$(x_1, x_2, x_3, x_4) = (1, 2, 3, 4) \quad \underline{\underline{\text{Ans}}}$$